were entirely dependent on the skill of the ground operator, as no automated stabilization or feedback systems existed. Structurally, these drones were simple, lightweight, and often constructed from wood, fabric, or early metallic alloys, resembling small fixed-wing aircraft.

One of the earliest known examples was the “Aerial Target” (1916), developed by A. M. Low in the United Kingdom, which demonstrated the feasibility of radio-controlled flight. This was followed by the “Queen Bee” (1935), a radio-controlled target drone used by the Royal Navy for anti-aircraft gunnery training. The term “drone” itself is believed to have originated from this model, inspired by the sound produced by its engine, akin to that of a male bee.

Although their technological capabilities were limited, first-generation manual drones laid the foundation for modern unmanned flight. They introduced the essential principles of remote piloting, control mechanisms, and system integration that would later evolve into advanced autonomous systems. This generation marked the beginning of a transformative journey – from human-dependent remote aircraft to the intelligent, automated drones that define contemporary aerospace innovation.

Figure 1.3 illustrates the structure and fundamental components of a first-generation manual drone, representing the earliest stage of unmanned aerial vehicle (UAV) development. These drones were primarily used during the early to mid-20th century for military purposes such as target practice, reconnaissance, and pilot training. The figure highlights key components, including the propeller, which provides thrust for forward motion; the fuse-lage, which houses the engine and basic control systems; the landing gear, which supports takeoff and landing; the radio antenna, responsible for receiv-ing manual control signals from the ground operator; and the tail assem-bly, which ensures flight stability and directional control.